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Fostering resilience of prospective startup businesses: a longitudinal study

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ABSTRACT

The growth of startup businesses has received public attention worldwide. Incubation and acceleration programs initiated by universities, government agencies, and companies stimulate movements in the startup ecosystem to improve the success rate of the prospective startup. This paper aims to explain how business incubators nurture potential startups as new business ventures in a dynamic environment by relying on the organizational resilience theory. The study adopts action research using a case study approach involving a university-based incubation in Indonesia during 2014–2024. The main contribution of this study lies in the organization's commitment to increasing the capacity of startups to succeed and strengthening the entrepreneurial ecosystem in the region following the three-phase framework: pre-incubation, incubation, and acceleration as a future reference for startup sustainability. The qualitative assessment of the study focuses on exploring affectionate elements: founders' commitment to sustainability, teamwork solidity, and mentoring effectiveness by the incubator as promoter and catalyst of startup survival and resilience. The study critically reflects on startup founders' behavioral changes, especially post-incubation. The phenomena indicate the variability of entrepreneurial orientation and mentality of a startup's success or failure. Active contributions and fruitful synergies among parties, such as startup talents or founders, facilitators, mentors, angels, and ventures are essential for startups' resilience and stakeholders' expectations. The findings also provide insights into managing technological innovations and policy developments in startup businesses.

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1. Introduction

Published articles on startup businesses have received attention from innovation management and public policy activists. Much attention is paid to factors affecting start-up success or failure, but little is paid to building resilience in a dynamic environment (Hossain & Valliere, 2024; Hung Kee & Rahman, 2017; Silva et al., 2023). A recent study shows a positive relationship between the individual resilience of entrepreneurs and the resilient response of small and medium-sized enterprises (SMEs) mediated by organizational resilience (Leonelli et al., 2025). Apart from that, the dichotomy of developing startup businesses and micro, small, and medium-sized enterprises often creates confusion among incubators in several aspects, such as prospective talents, innovation capacity, delivery methods, and scaling up of the business (Del Sarto et al., 2021). This confusion indicates concern about finding a breakthrough model for building an entrepreneurship climate and increasing concern about delivering innovative solutions to society.

Universities, industries, and government agencies run startup incubation programs to boost the number of young entrepreneurs in the country. Startup incubators traditionally provide coaching and mentoring in several key aspects of a successful business. The success stories of technopreneurs who rely on digital innovation spur young people to initiate startups as new ventures in certain industries (Salamzadeh & Kirby, 2017). However, new ventures sometimes overlook the behavioral complexity of dealing with the

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resilience of the business (Leonelli et al., 2025). They missed key points in developing entrepreneurial orientation and the mindset of prospective startups as a mediating variable for successful new ventures in the industry (Hung Kee & Rahman, 2017). The question is how potential startup talents are nurtured and developed during the incubation and acceleration.

As intermediaries, incubators and accelerators are critical in developing an entrepreneurial mindset and strengthening startups' resilience. Understanding the inherent traits of startup entrepreneurs mediated by their past experiences consistently influences startups' resilience (Branca et al., 2025). Their capacity and capability to deal with uncertainty are the main reasons. In that situation, founders' and co-founders' consciousness and emotional stability become controlling factors in the decision-making and business sustainability.

Relevance

Diverse approaches to running the incubation programs are currently being put in place, which indicates that startup development is still evolving, starting from ideation, proof of concept, incubation, and acceleration to finally spin-offs. Startup growth reflects the success of talent development and entrepreneurship programs. Dutta et al. (2024) explain how the innovation index varies across countries depending on the readiness level of the institution, human capital, infrastructure, market sophistication, and business sophistication. Consequently, the approach to startup development and the impact of innovation also vary.

Lee et al. (2017) described how the startup business ecosystem was evolving in Korea, China, and Japan. Each country showed very different characteristics of a start-up ecosystem, i.e. Korea has a supply-centric ecosystem, but Japan and China have a technology-centric and a market-centric ecosystem, respectively. Meanwhile, Greco (2025) describes unique patterns of high-performing startups facilitated by the startup-friendly ecosystem predominantly clustered in Central and Southern Italy. Therefore, a clear strategy for nurturing young potential talents became a crucial aspect of building the strong character of startup founders facing the creative destruction phenomenon (Gans et al., 2002). The astuteness of startup founders in exploring opportunities for value creation was part of an opening asset for surviving and sustaining businesses in their country.

Empirical evidence showed that the development of startups with temporal and spatial uniqueness in tough conditions benefits more from their wisdom in interpreting the existing environment (Hong et al., 2015). Managing the proximity of access to innovation resources and strengthening networks within a spatial environment offered value-creating opportunities for startup development. The proximity of access to sources of innovation allows businesses to gain first-mover advantages before their creativity is followed by others (von Hippel, 1994). Later, Eggers et al. (2017) stated that critical thinking spurs creativity and innovation, which mediates business performance. Prospective startups gained the benefits of critical thinking through mentorship and expert facilitation during their incubation and acceleration processes (Cohen et al., 2019; García-Ochoa et al., 2022). These phenomena imply that physical proximity and social interaction stimulate startup entrepreneurs' consciousness to build closeness with expert facilitators, mentors, funders, and communities within startup ecosystems.

Contextual settings

This article explores how the incubation program run by a public university fosters the resilience of prospective startups in a dynamic environment. The program called *Innovative Academy* was founded in 2014 at Universitas Gadjah Mada (UGM) Indonesia to build startups and deliver innovative solutions to society's problems. During 10 years (2014–2024), *Innovative Academy* has fostered 295 prospective startups through incubation and acceleration activities. The academy adopts a semi-structured approach to provide a venue for young talented candidates to be creative and innovative starting from ideation, proof of concepts, prototyping, and then making it happen. Not all startups were finally able to survive and then grow their business for various reasons, such as personal matters of founders, team members' fragility, and the effectiveness of the mentorship processes.

Historically, back in 2014, *Innovative Academy* was a kind of alternative extra-curricular program for entrepreneurship development on campus. The program proposes that entrepreneurial orientation can

be built through semi-structured activities so that they do not rely entirely on inborn traits. The talent search, selection, and mentoring process was carried out by a dedicated team formed by the university. The participants initially were undergraduate and graduate students who had taken several semesters and acquired basic knowledge on campus. The program succeeded in developing several startups with a pool of talent and idealism to solve community problems through technology-based innovation.

Over time, student startups' survival rates were quite low due to their decision to pursue further careers as professional workers. Pressure from close family members and self-orientation to obtain a regular income for living became the reasons for their decisions to leave startups. An unexpected reason is economic rationality and pragmatic orientation to survive in their initial career. As a startup builder and incubator, Innovative Academy then opened a space for relatively mature talents, including graduate students, alumni, and startup activists, to join the program.

In line with developments, the university's incubation program was modified to involve industry and government institutions as co-promoters in strengthening the startup ecosystem. New business models emerged. Talents were supplied with real problems in the industry. The involvement of industry partners in the incubation process was one way to resolve the bank of problems in a relatively limited time. The research question is how incubators can guide the startups they foster to survive and be sustainable in upcoming events. How can industry involvement impact the survival rate of emerging start-ups? The answers will significantly build startup resilience and contribute to knowledge development as a precursor for practical implications in entrepreneurship and innovation.

Meanwhile, at the macro level, several ministries in Indonesia (Ministry of Communication and Digital, Ministry of Industry, Ministry of Higher Education, Science, and Technology, Ministry of Tourism, and Ministry of Cooperatives and Small and Medium Enterprises) have initiated programs to encourage the growth of startup towards a digital economy. Moreover, several state-owned enterprises have also taken the initiative independently or in collaboration with universities to recruit potential talent and run startup incubations. These initiatives brought a new signal of urgency to move forward to a stepping stone in overseeing the innovation process in the digital era and the implementation of the entrepreneurial spirit for national productivity.

Finally, the remainder of this article is structured as follows: the next section briefly examines the literature on factors contributing to startup resilience. Then, the following section presents Action Research as a qualitative research approach to exploring the resilience and sustainability of startups in different industries, followed by the findings of this empirical work. The conclusions and implications of this study are briefly presented at the end of the article.

2. Literature review

As mentioned in the introduction, published articles regarding startup success or failure were quite diverse. Empirical studies focused on developing the resilience of start-up businesses, especially in facing changes due to volatile, uncertain, complex, and agile environments, were also relatively limited. According to Hamel and Välikangas (2003), organizational resilience refers to the capacity for continuous reconstruction. It requires innovation concerning those organizational values, processes, and behaviors that systematically favor perpetuation over innovation. Later, Vogus and Sutcliffe (2007) defined resilience as the maintenance of positive adjustment under challenging conditions such that the organization emerges strengthened and more resourceful. Table 1 summarizes contributing variables to startup resilience.

Organizational resilience is a multi-level concept related to the dynamic capabilities of organizational resources, routines, and processes (Schoemaker et al., 2018). It is also passed through individuals, groups, and organizations, depending on the level of interactions among different contexts (Xiao & Cao, 2017). Following this argument, the resilience of a start-up business is formed from a dynamic process involving cognitive, emotional, and relational aspects, enabling the organization to overcome unexpected challenges.

The founders' mentality of facing uncertainty with concrete action is a provision to survive, grow, and develop (Zook & Allen, 2016). Collective awareness of teamwork and managing organizational behavior in a conducive innovation ecosystem is relevant to startup development. The founders' readiness to leap

Table 1. Contributing variables to resilience.

Key variables	References	Remarks
Commitment to sustainability	Zook and Allen (2016), Picken (2017), Schötz (2025)	The founders' commitment to delivering results is the basis for consistent action for startup growth and sustainability.
Teamwork Solidity	Mitchell and Boyle (2019), O'Reilly and Binns (2019), Adamides et al. (2022), Hanifzadeh et al. (2024)	Solid teamwork and the alertness of team members as strategic arms for startups to perform in a dynamic environment.
Mentorship effectiveness	Hampel et al. (2020), García-Ochoa et al. (2022), Noronha and Pillai (2025)	Effective mentoring and resource flexibility in an open innovation climate is necessary for the startup acceleration process.

into the value creation process is part of risk mitigation against uncertainty. Therefore, building startup readiness and resilience is a long process involving key variables, including the founders' commitment to sustainability (Picken, 2017), teamwork solidity (Mitchell & Boyle, 2019), and mentoring effectiveness (Hampel et al., 2020).

2.1. Founders' commitment to sustainability

Since the beginning, von Hippel (1994) has described access to sources of innovation supports creativity and strategy development. The opportunity to gain competitive advantages can be obtained because of its proximity to sources of innovation. Spatial proximity to research and development centers opens opportunities for reciprocal collaboration between partners and a sense of novelty in strengthening the open innovation climate (Usman & Vanhaverbeke, 2017). An open innovation ecosystem facilitates external and internal ideas to advance organizational innovations through which dynamic capabilities boost knowledge exchange and the learning process to explore possibilities (Bogers et al., 2019). Among startup founders, access to new knowledge and different ways of thinking reflects the eagerness to deliver advantages and constructive meanings within their ecosystems.

Silva et al. (2023) investigate how startups' business processes were affected by a difficult situation such as the pandemic Covid-19 and how founders and co-founders managed this sudden change and challenge in an uncertain and complex environment to survive. Schötz (2025) argues that static resilience is closely related to survival probability considerations during disruptive situations, whilst integration with dynamic resilience proves the level achieved through the speed of recovery after the disruption. A startup's resilience emerged due to the founders' creativity and influence power of self-improvement in maintaining customer engagements and business connections with other stakeholders. Previous studies also showed that creative thinking fosters the innovation performance of startups, leading to concrete steps for sustainability (Eggers et al., 2017). The argument relies on the commitment of startup founders and co-founders to deliver results and consistent actions for resilience.

In addition, as prospective leaders, founders' sensitivity towards creativity can be placed as support elements for problem-solving and value creation, especially in virtual work (Humala, 2015). The founders' views, motivation, and ambition on the start-up business and how they observed virtuality will become natural prerequisites for work conditions and productivity. Opportunities for the adoption of innovation and value creation are obtained through commitment to building synergy and collaboration between stakeholders, compromising social logic and economic logic. As Rogers (1983) explained, the diffusion of innovation is a social process where ideas, practices, and innovations are delivered to different parts of society through communication networks. The diffusion theory is quite relevant to explaining why and how internal organizations explore communication channels to build solid teamwork.

2.2. Teamwork solidity

Solid teamwork and the alertness of team members as strategic arms for startups to perform in a dynamic environment. The argument relies on the belief that strengthening the capacity of start-ups to implement innovation strategies is crucial in facing uncertain conditions. Building resilience with high flexibility is needed to respond to the complexity and volatility of startup development. However, the

time length it takes for an innovation launched by a startup business to pass through the Death Valley of innovation continues to invite academic debate.

Founders' encouragement to maintain their strong achievement emerged, although the readiness stage of the startup may vary. O'Reilly and Binns (2019) explained that the path of disruptive innovation for startups goes through three stages: ideation to make breakthroughs in solving problems; incubation to validate ideas in real situations; and scale-up by reallocating assets and capabilities needed for growth. Successful startups are usually being able to go through all of these stages accordingly.

Adamides et al. (2022) explained how a group of innovators becomes an entrepreneurial team through the transformation of its work practices. They argued that prospective startups benefit from business accelerator programs by deploying specific practices and routines. Business accelerators usually help startups acquire, integrate, administer, and reconfigure their endowment of resources to deal with the dynamic environment.

Apart from collective founders and co-founders, solid teamwork in startup development was the key to survival, growth, and sustainability. Stayton and Mangematin (2016) argued that continued public support for startup businesses was in the interests of many people for some reasons: accelerating innovation, employment opportunities, and the impact on economic prosperity. From a resource optimization perspective, achieving these three startup businesses required managing behavioral aspects based on new values an adaptive organizational culture, and solid teamwork. Later on, Hanifzadeh et al., (2024) explain that the combination of entrepreneurial teams including team-level cognitive, comprehensiveness, team commitment, team size, and team integrity affects the scalability of startups.

2.3. Mentorship effectiveness

Every startup is unique. Not all startups need the same amount of support from incubators to thrive in the market (Máté et al., 2024). For example, startups engaged in manufacturing require more support and longer incubation times than startups in software development. García-Ochoa et al. (2022) explained how business incubators and accelerators extrapolate resources to build dynamic capabilities within startups. The results of their study provide answers to the need for effective mentoring services. Nonetheless, it is unknown what services are most helpful for startups in each type of incubator and accelerator.

Entrepreneurial activities in incubators are quite diverse along the journey, although there are similarities in the patterns developed by each, starting from meetups, ideation, prototyping, workshops, demo day, pitching, and finally spinning offs. There is no single formula that can be applied successfully to all circumstances. Page and Holmström (2023) mention that some aspects of incubators' experience in dealing with professional coaching and mentoring services to their tenants are crucial, such as market research, financial management, cost accounting, Intellectual Property protection, and preparing legal contracts. Other services managed by incubators include motivation for self-development, presentation skills, business plan preparation, investor relations, and some thematic workshops are also essential. All of these services are not mandatory but are provided as needed.

Meanwhile, providing services such as co-working space for startups, reliable Internet access, amenities, and strengthening business networks are considered supporting functions for incubation and acceleration activities. An intermediation function for startups to build interaction and business networks. Orchestration of resources to achieve shared goals with stakeholders fosters a new climate and culture of innovation amongst startup businesses. Therefore, the success of an incubator in guiding start-up businesses is largely determined by a combination of various factors, both due to the uniqueness or differences of incubators and the reliability of the services provided in their ecosystem.

In addition, Hampel et al. (2020) argued that new ventures are required to identify and manage their relationships with key stakeholders to find the right direction for the business. A pivot action in their business models or even value propositions might be extrapolated to accommodate possible gain and pain gaps coming out from the proposed ventures. Pivot is a change process carried out due to adjustments as a consequence of the dynamics of environmental conditions. This suggests that an intensive coaching and mentoring process during the incubation or acceleration is a necessity for prospective startup businesses.

At last, the presence of dedicated startup incubators contributes to the formation of entrepreneurial behavior and mentality among talented candidates. Noronha and Pillai (2025) are also concerned with fostering startups through integrating entrepreneurship practices and business incubation and acceleration programs to spur economic development. The openness and institutional facilitation atmosphere stimulate participants to share knowledge, build solid teamwork, and develop new business ventures. Battistella et al. (2017) emphasized that the open innovation climate developed during incubation plays a key role in realizing startup success. Therefore, entrepreneurial spirit was introduced to young talented candidates to bring new optimism based on the power of digital innovation. Access to external knowledge and business networks were considered critical factors in overcoming innovation obstacles among startups.

3. Research method

This exploratory study relies on Action Research methodology to explore the behavioral changes of startup businesses facilitated by *Innovative Academy*, the campus-based incubator, as an object of the study. Following Yin (2018), the *Innovative Academy* was chosen as a case study to answer the question of how the startup incubator manages its programs and activities toward resilience. Specifically, it was motivated by the need to deepen the understanding of the incubation and acceleration processes that led to startup success or failure and identify critical factors.

Researchers are directly involved in the processes, simultaneously codifying data for analysis and answering the research question. As stated earlier, during 10 years (2014–2024), *Innovative Academy* has fostered 295 prospective startups through pre-incubation, incubation, and acceleration activities. Startup performance is evaluated at every stage to seek eligibility for subsequent processes. This longitudinal field study involves the participation of prospective start-up businesses, mentors, and administrators under the coordination of the Innovative Academy program, a campus-based incubation at Universitas Gadjah Mada, Yogyakarta, Indonesia.

The action research in this study was carried out in three critical phases. In the *first* phase, potential talents were selected from final-year students of the University through hackathons. It involves 295 prospective startups in the program from 2014 to 2024. The activity focused on pre-incubation to develop ideas and proof of concept. The prospective startups were asked to identify real society's problems and enable technological innovation to find the best solution. In the *second* phase, the activities focused on the validation incubation process to ensure that innovation prototypes, business models, market penetration, and revenue streams run as planned. Participants in this second phase came from pre-incubation alumni and new participants who already had innovations ready and worthy of incubation. In the *third* stage, the acceleration and scaling-up process were carried out for potential startups. Not all startups in the incubation process make it to the acceleration stage for various reasons. Therefore, startups that reach the acceleration stage have the potential to grow and maintain their capability and chance for sustainability of their businesses.

Direct involvement and observations of researchers in all stages were carried out from the pre-incubation, incubation, and acceleration leading to spin-offs. Activity notes and photovoice documentation were used as supporting data to interpret the development of startups and the problems they face. At each phase, potential startups can pivot or make changes to their business model, management team, and target market depending on the uniqueness of the problems faced by the startup. The series of discussions was then selected based on codified data to gain an in-depth understanding of the experience of managing incubation and acceleration processes. The activities mainly oversee tenant progress and monitor prospective startups after they complete the program. Propositions regarding the formation of resilience and attribution of added value process raised by the incubator are the material for discussion. The process flow for strengthening startup resilience is schematically shown in Figure 1.

4. Analysis and discussions

As a pioneer in developing a campus-based startup ecosystem in Indonesia, *Innovative Academy* seeks to find the ideal startup incubator format, but of course, it faces various challenges. The initial idea is that

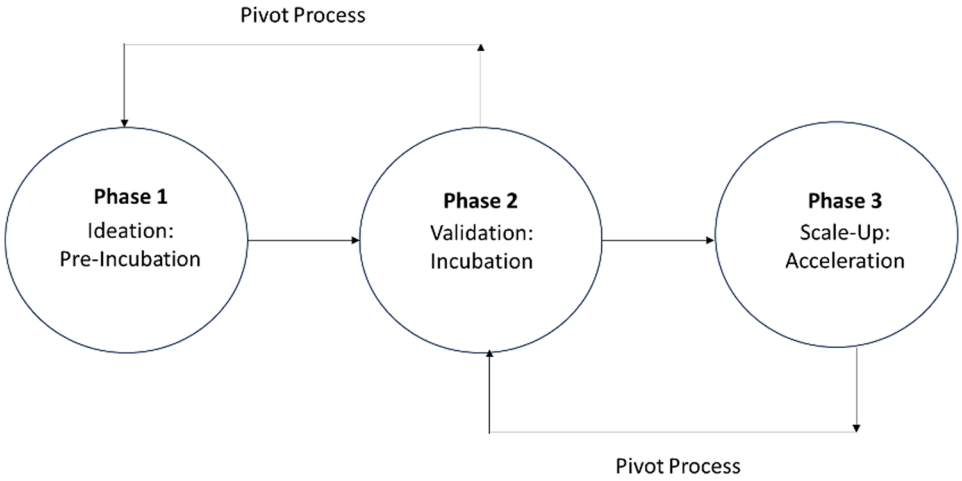

Figure 1. Strengthening process of startup resilience.

the young generation has the energy to solve various national problems with digital technology and innovation. Potential talents from students, alumni, and even communities with the same vision are open to strengthening the region's startup and entrepreneurship ecosystem.

Every startup was encouraged to explore digital technology as an enabling factor for business operations and productivity. The composition of talents in each startup was attempted to balance engineering and non-engineering competencies. This process design aims to optimize the capacity and capability of prospective talents to generate technology-based innovation solutions. A combination of technical and social approaches was exercised to get deep tech competency and the behavioral aspects of innovation.

Direct observation was conducted to identify problems and challenges in the program as suggested in the case study method (Yin, 2018). Each startup must follow the standard rules in line with the startup development process starting from pre-incubation, incubation, and acceleration. There were no exceptions for participants from various fields to explore potential and utilize the availability of reliable sources of innovation (von Hippel 1994). The *Innovative Academy* performed as a catalyst to accelerate startups' capacity to find problems and innovative solutions. The indicator of success was shown by the number of participants who consistently managed activities along the process. Table 2 presents the number of startups that have completed the pre-incubation, incubation, and acceleration stages with their existing active status. Some startups enrolled in the *Innovative Academy* (IA) program in collaboration with strategic partners (noted as "IA-affiliated").

During a decade of the academy (2014–2024), at least three phases came out in the development process among startup talents. *First*, the pre-incubation phase, mainly built startup literacy and the founders' commitment to delivering results (Picken, 2017; Zook & Allen, 2016). *Second*, the incubation phase strengthened team members' solidity through a collective awareness of business dynamics (Mitchell & Boyle, 2019; O'Reilly & Binns, 2019; Adamides et al., 2022). *Third*, the acceleration phase focuses on the effectiveness of the mentoring process for startups (García-Ochoa et al., 2022; Hampel et al., 2020). These three milestones or phases go hand in hand with the euphoria of startup business in Indonesia which was promoted by government institutions, universities, and private corporations according to their respective roles.

Each promoter (government, university, and corporation) played a strategic role in delivering startups with all their uniqueness. Figure 2 presents the quantification of support from each promoter in attracting the number of participants through the hackathon process, the number of start-up teams formed, and the current status of the start-up. The high-low level of promoter quantification describes the involvement of promoters in startup development. When promoters only come from universities, startup achievements are limited as depicted in the first 2 years of activities. Entering the third year, the industry involvement as promoters began to appear reflecting the growth in the number of startups. Furthermore, when the *Innovative Academy* at the university collaborated with industry and government as depicted

Table 2. Number of startups and existing status.

Year	Startups (IA)	Pre-incubation	Incubation	Acceleration	Existing
2014	3	3	3	2	0
2015	14	14	14	2	2
2016	18	24[a]	24	4	1
2017	9	9	9	2	1
2018	42	44[b]	44	6	8
2019	15	16[c]	16	0	6
2020	24	30[d]	30	0	15
2021	67	73[e]	73	14	73
2022	20	20	20	0	20
2023	45	49[f]	49	14	49
2024	38	38	38	25	38
Total	295				

[a,d,e]6 IA affiliated, [b]2 IA affiliated, [c]1 IA affiliated, [f]4 IA affiliated.
Source: Primary data.

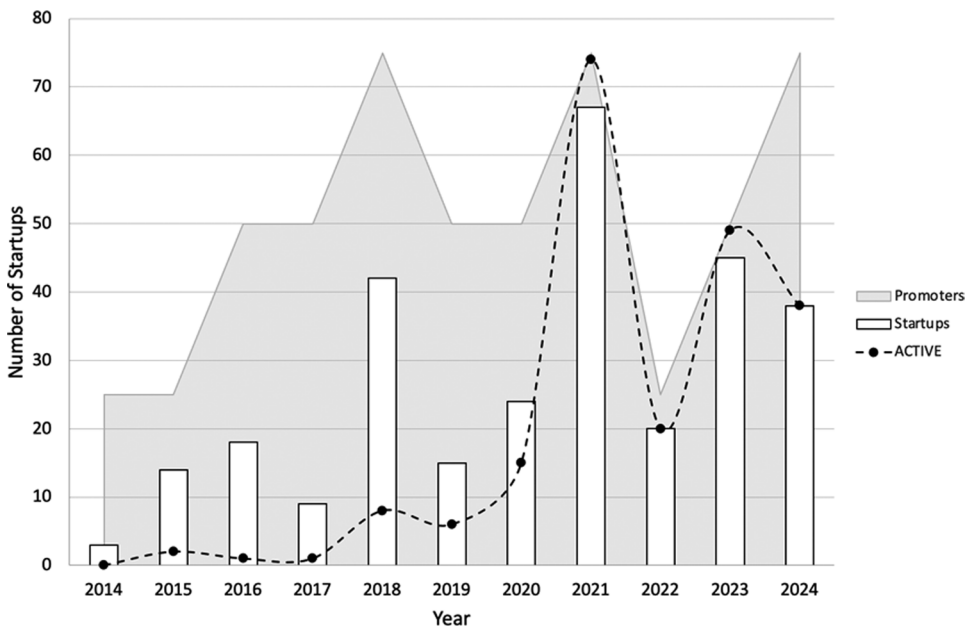

Figure 2. Quantification support from promoters, number of startups, and current status.

in 2018, 2021, and 2023, the number of startups grew substantially. This finding was consistent with Leydesdorff and Etzkowitz (1998) regarding the triple helix relationship (university, business, and government) in strengthening and accelerating innovation.

4.1. Phase 1: startup literacy and commitment to business sustainability

In the first 2 years of the activities (2014–2016), the University became the main promoter by attracting students to join this regional entrepreneurship program. The business incubator on campus provided co-working space and facilities as supporting infrastructure and qualified mentors as volunteers to build ecosystems. As a pioneer in the area, the realistic target was to increase startup literacy during the pre-incubation stage by offering a different scheme from the ordinary development scheme for small and medium-sized enterprises. The program finally called for participants mostly final-year students from various faculties and vocational schools. Series workshops and continued hackathons were conducted to attract potential talents and to build teams.

The teams formed were required to be multidisciplinary, a combination of engineering, health, agriculture, business, and social sciences. Potential talents in multidisciplinary teams were a strength but a challenge if not supported by a qualified mentor. They were asked to explore ideas and formulate them into a proof of concept that provided solutions to problems faced by society at that time. The biggest

challenge is building commitment to sustainability among founders and co-founders due to limited research skills, fragile business models, and unclear value propositions related to the problem and innovative solutions offered. As a result, out of 35 startups formed over 2 years, only three startups survived until the spin-off stage.

However, new conditions changed quite significantly. Startups formed in the following years (2017–2020) experienced a slightly different situation due to incoming strategic industry partners in building start-up literacy and strengthening the ecosystem. Industry partners were actively involved in hackathons, pitching potential candidates, and mentoring processes together with internal resources from the University. The combination of academia and industry later became a model of campus-based startups. The finding showed the helix logic for understanding and speeding up innovation and learning processes, especially in startup development (Leydesdorff & Etzkowitz, 1998; Miron, 2008).

The double helix model brought a new spirit and optimism among participants. Active status in the program increased due to the addition of new participants who entered directly into the incubation stage or previous participants who continued to deepen through the pivot process. The success rate up to the level of spinoff also increased in line with the commitment of stakeholders from the industry, despite the large number of startups that were ultimately considered to have failed during the process. Most startup candidates fell out during the nurturing process for various reasons: pursuing a career elsewhere as a professional, being disbanded, changing business models, not getting suitable funding partners, and feeling they could not survive with a startup.

4.2. Phase 2: building a solid team through incubation

When defining the success of a startup business, public attention is focused on how the financial bottom line grows and the size of the business valuation. Meanwhile, the startup foundation that relies on non-financial aspects, especially the capacity of human resources and the motivation of the founders as a focal point to maintain the existence and sustainability of the business is neglected during the incubation (Adamides & Karfaki, 2022). Moreover, when the 2018 startup euphoria also occurred in Indonesia following the global trend, the public expectation of startups was quite high. The existence of startups was highlighted through inorganic growth patterns and exponential valuations. Startup growth attracted many observers and investors, ultimately giving rise to the startup bubble phenomenon and raising the valuation of startup businesses.

The massive campaign about startups by the Ministry of Communication and Information, the Ministry of Cooperatives and Small and Medium Enterprises, and the Ministry of Industry Republic of Indonesia boosted the enthusiasm of young talents to participate in campus-based startup business programs. The dream of the founders and co-founders to make their startups into Unicorns continued until entering the Covid-19 Pandemic which boosted various digital innovation platforms. The massive campaign was perceived as an opportunity to strengthen the entrepreneurship ecosystem with support from industry and government in parallel running incubation programs on campus.

On the other hand, the expectations of founders and co-founders to be able to grow in a short time with the support of global venture capital funding raise behavioral challenges in business management. Understanding good business practices and prudent or good corporate governance requires self-discipline and the entire startup team. Cost-conscious behavior and optimizing resource capacity so that business operations run smoothly and continuously are instilled in the minds of founders and startup managers. Ironically, in overseeing the incubation process, we still found the behavior of founders and co-founders who try to harvest their investments in startups without considering the efficiency and sustainability of business operations.

The dominant role of a founder or co-founder in operational activities makes startups fragile in terms of governance mechanisms, especially regarding the use of public funds that have been invested in startups. The solidity of the management team in implementing a performance-based strategy became a necessary condition for organizational resilience (Xiao & Cao, 2017). A solid management team was also needed for startups facing dynamic business environment conditions. As O'Reilly and Binns (2019) mentioned, the ability of startups to go through critical stages of disruptive innovation was crucial. Therefore, cultivating startup resilience was the key to maintaining a new balance in the innovation ecosystem.

An illustrative startup is Inspirasien which can be seen at https://www.inspirasien.id/ (Accessed 25 March 2025). This startup focuses on innovation in a health support platform and patient assistance including patient services, caregivers, communities, and institutions that care about a better quality of life. Managing business models and revenue streams becomes a major challenge for this startup passing through the critical stage during the incubation and acceleration process. This implies team solidity during the incubation process will strengthen the capacity of the startup to survive and sustain in a dynamic environment.

Meanwhile, startups that finally survive passing the incubation stage have the resilience to go through difficult times like those faced by their peers. Based on the founders' self-expression and commitment to maintaining the startup's success, there was always an opportunity for continuous improvements. This study also found that the time spent and the readiness of the startup upon completion of incubation were important determinants of whether or not the startup succeeded after leaving the incubator's safety net.

4.3. Phase 3: acceleration and mentoring effectiveness

During the COVID-19 pandemic, incubation and acceleration activities in the *Innovative Academy* program were relatively limited due to government policy on physical and social distancing. The incubation and acceleration programs and activities were carried out entirely online. Even so, their enthusiasm and efforts in participating actively in the series of activities remained intact. In fact, during difficult times during the pandemic, creativity and innovation emerged from startups. As García-Ochoa et al. (2022) explained, business incubators and accelerators might extrapolate resources to build dynamic capabilities within startups.

An illustration of a startup case study in this case is Stechhoq Robotika Indonesia which grows and develops in the area of robotics innovations for medical devices, education, and industrial purposes. Detailed descriptions of Stechoq's activities can be seen on the following web page https://stechoq.com/ (Accessed 25 March 2025). Another startup case study illustration is Helber, which focuses on moving into the electronics sector for appropriate industries, such as the application of IoT, medical devices, Artificial Intelligence, and educational technology. Details of Helber can be found at the following web page https://helber.co.id/ (Accessed 25 March 2025). Both illustrative cases reflect the effectiveness of mentoring and monitoring of the prospective startups, which then provides a positive signal regarding the incubation and acceleration process carried out in the program.

The commitment of *Innovative Academy* management in providing services and helping to oversee the sustainability of startups is demonstrated by continuing to facilitate the availability of reliable mentors according to their respective competencies. Moreover, the flexibility of access and connection with members of the startup network is also widely open. The principle of open innovation is applied at *Innovative Academy* and makes it continue to grow as a comfortable second home for startups. The emotional bond formed between mentors at Innovative Academy and founders is social capital to maintain the performance and sustainability of startups.

The social interaction during the acceleration phase was quite strong, especially after the pandemic. The COVID-19 pandemic has made startup entrepreneurs aware of mutual dependency by fostering a spirit of cooperation in sustainable innovation. The ambiance and warmer relationship create enlightening moments for all parties. The management of the *Innovative Academy* program can now more effectively monitor the progress of each startup being mentored because they provide mutual support to achieve their respective performance targets. In such a situation, the effectiveness of the mentoring process can be an important milestone in building the resilience and sustainability of startups. This has been empirically proven in the incubation and acceleration process with a strong effort to accelerate the growth of campus-based startups. It is a promising endeavor in developing innovation-driven enterprises and entrepreneurship ecosystems.

5. Conclusions and implications

This exploratory study reveals key elements regarding the fostering resilience issues of startup businesses. The major point was the urgency of strengthening behavioral aspects in the startup development

starting from pre-incubation, incubation, and acceleration processes. The resilience of startup entrepreneurs through the behavioral process is a robust initiative and an optimistic signal for business sustainability. The qualitative assessment of the study explores the three affectionate elements, namely, founders' commitment to sustainability, teamwork solidity, and mentoring effectiveness, by the incubators as promoters and catalysts of the startup business. A perfect combination of these elements is essential for start-up survival and resilience when facing uncertainty.

The study also reveals critical reflection on behavioral changes amongst founders, especially post-incubation. The phenomena indicate the variability of entrepreneurial orientation and mentality of startups' survival and sustainability. The findings provide insights into managing technological innovations and policy developments in startup businesses. Public funding policies on startups should ideally be linked to the agenda of strengthening the foundation and capacity of startups to operate in a dynamic business environment. Finally, the proximity of startups to sources of creativity and innovation in universities can be a basic foothold in strengthening the entrepreneurial ecosystem at the national level.

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Author contributions

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Data availability statement

The data supporting this study's findings are available on request from the corresponding author, Hargo Utomo. However, due to research institution policy, the data are not publicly available.

References

Adamides, D. E., & Karfaki, E. (2022). Transitional practices in the pre-start-up phase of academic spin-off creation: An explanatory case study. *International Journal of Entrepreneurship and Innovation Management*, *26*(1/2), 1. https://doi.org/10.1504/IJEIM.2022.120878

Battistella, C., De Toni, F. A., & Pessot, E. (2017). Open accelerators for start-ups success: A case study. *European Journal of Innovation Management*, *20*(1), 80–111. https://doi.org/10.1108/EJIM-10-2015-0113

Bogers, M., Chesbrough, H., Heaton, S., & Teece, J. T. (2019). Strategic management of open innovation: A dynamic capabilities perspective. *California Management Review*, *62*(1), 77–94. https://doi.org/10.1177/0008125619885150

Branca, E., Intenza, M., & Doni, F. (2025). Startup entrepreneurs' personality traits and resilience: Unveiling the interplay of prior experience. *International Entrepreneurship and Management Journal*, *21*(1). https://doi.org/10.1007/s11365-024-01023-y

Cohen, L. S., Bingham, B. C., & Hallen, L. B. (2019). The role of accelerator designs in mitigating bounded rationality in new ventures. *Administrative Science Quarterly*, *64*(4), 810–854. https://doi.org/10.1177/0001839218782131

Del Sarto, N., Di Minin, A., Ferrigno, G., & Piccaluga, A. (2021). Born global and well educated: Start-up survival through fuzzy set analysis. *Small Business Economics*, *56*(4), 1405–1423. https://doi.org/10.1007/s11187-019-00238-6

Dutta, S., Lanvin, B., Leon, R. L., & Wunsch-Vincent, S. (2024). *Global Innovation Index 2024: Unlocking the Promise of Social Entrepreneurship* (17th ed.). Word Intellectual Property Organization (WIPO).

Eggers, F., Lovelace, J. K., & Kraft, F. (2017). Fostering creativity through critical thinking: The case of business startup simulations. *Creativity and Innovation Management*, *26*(3), 266–276. https://doi.org/10.1111/caim.12225

Gans, S. J., Hsu, H. D., & Stern, S. (2002). When does start-up innovation spur the gale of creative destruction? *The RAND Journal of Economics*, *33*(4), 571–586. https://doi.org/10.2307/3087475

García-Ochoa, P. C., Heredero, D. C., & Jimenez, B. J. F. (2022). The role of business accelerators in generating dynamic capabilities within startups. *International Journal of Entrepreneurship and Innovation Management*, *26*(1/2), 25. https://doi.org/10.1504/IJEIM.2022.120879

Greco, F. (2025). Overseas benchmarking for startup ecosystems in Central and Southern Italy. *International Entrepreneurship and Management Journal*, *21*(1). https://doi.org/10.1007/s11365-024-01028-7

Hamel, G., & Välikangas, L. (2003). The quest for resilience. *Harvard Business Review*, *81*(9), 52–63, 131.

Hampel, E. C., Tracey, P., & Weber, K. (2020). The art of the pivot: How new ventures manage identification relationships with stakeholders as they change direction. *Academy of Management Journal*, *63*(2), 440–471. https://doi.org/10.5465/amj.2017.0460

Hanifzadeh, F., Talebi, K., & Jafari-Sadeghi, V. (2024). Scalability of startups: The impact of entrepreneurial teams. *Journal of Global Entrepreneurship Research*, *14*(1). https://doi.org/10.1007/s40497-024-00383-7

Hong, E., Lee, H. I., Sun, L., & Harrison, R. (2015). Entrepreneurship across time and space: Empirical evidence from Korea. *Small Business Economics*, *44*(3), 705–719. https://doi.org/10.1007/s11187-014-9613-1

Hossain, M. S., & Valliere, D. (2024). Emotional transformation of failure: Passion, resilience and entrepreneurial self-efficacy. *Journal of Developmental Entrepreneurship*, *29*(04), 2450022. https://doi.org/10.1142/S1084946724500225

Humala, A. I. (2015). Leadership toward creativity in virtual work in a start-up context. *Journal of Workplace Learning*, *27*(6), 426–441. https://doi.org/10.1108/JWL-08-2014-0059

Hung Kee, D. M., & Rahman, A. M. (2017). Analyzing entrepreneurial orientation impact on start-up success with support service as moderator: A PLS-SEM approach. *Business and Economic Horizons*, *13*(2), 128–141. https://doi.org/10.15208/beh.2017.10

Lee, M., Lee, M., & Kim, J. (2017). A dynamic approach to the start-up business ecosystem: A cross-comparison of Korea, China, and Japan. *Asian Academy of Management Journal*, *22*(2), 157–184.

Leonelli, S., Campagnolo, D., & Gianecchini, M. (2025). Entrepreneur and organizational resilience: A multilevel perspective on Italian SMEs. *Journal of Small Business Management*, *63*(2), 757–785. https://doi.org/10.1080/00472778.2024.2351483

Leydesdorff, L., & Etzkowitz, H. (1998). The Triple Helix as a model for innovation studies. *Science and Public Policy*, *25*(3), 195–203.

Máté, D., Estiyanti, N. M., & Novotny, A. (2024). How to support innovative small firms? Bibliometric analysis and visualization of start-up incubation. *Journal of Innovation and Entrepreneurship*, *13*(1). https://doi.org/10.1186/s13731-024-00361-z

Miron, D. (2008). Linking the double helix of learning and work to the Triple helix of university – industry – government in the Europe of knowledge. *Management & Marketing*, *3*(4), 3–20.

Mitchell, R., & Boyle, B. (2019). Inspirational leadership, positive mood, and team innovation: A moderated mediation investigation into the pivotal role of professional salience. *Human Resource Management*, *58*(3), 269–283. https://doi.org/10.1002/hrm.21951

Noronha, L. C., & Pillai, B. V. K. S. (2025). Fostering startups through the vehicle of entrepreneurial support organisations like technology business incubators (TBIs): A systematic literature review. *Journal of Global Entrepreneurship Research*, *15*(1). https://doi.org/10.1007/s40497-024-00419-y

O'Reilly, C., & Binns, J. M. A. (2019). The three stages of disruptive innovation: Idea generation, incubation, and scaling. *California Management Review*, *61*(3), 49–71. https://doi.org/10.1177/0008125619841878

Page, A., & Holmström, J. (2023). Enablers and inhibitors of digital startup evolution: A multi-case study of Swedish business incubators. *Journal of Innovation and Entrepreneurship*, *12*(1), 35. https://doi.org/10.1186/s13731-023-00306-y

Picken, C. J. (2017). From startup to scalable enterprise: Laying the foundation. *Business Horizons*, *60*(5), 587–595. https://doi.org/10.1016/j.bushor.2017.05.002

Rogers, E. M. (1983). *Diffusion of innovation* (3rd ed.). The Free Press.

Salamzadeh, A., & Kirby, A. D. (2017). New venture creation: How start-ups grow?. *AD-Minister*, No. 30 January–June. https://doi.org/10.17230/ad-minister.30.1

Schoemaker, P. J. H., Heaton, S., & Teece, D. (2018). Innovation, dynamic capabilities, and leadership. *California Management Review*, *61*(1), 15–42. https://doi.org/10.1177/0008125618790246

Schötz, E. (2025). Resilience vs. survival: Same song, new melody? *Future Business Journal*, *11*(1). https://doi.org/10.1186/s43093-025-00427-8

Silva, E., Beirão, G., & Torres, A. (2023). How startups and entrepreneurs survived in times of pandemic crisis: Implications and challenges for managing uncertainty. *Journal of Small Business Strategy*, *33*(1). https://doi.org/10.53703/001c.72084

Stayton, J., & Mangematin, V. (2016). Startup time, innovation and organizational emergence: A study of USA-based international technology ventures. *Journal of International Entrepreneurship*, *14*(3), 373–409. https://doi.org/10.1007/s10843-016-0183-y

Usman, M., & Vanhaverbeke, W. (2017). How start-ups successfully organize and manage open innovation with large companies. *European Journal of Innovation Management*, *20*(1), 171–186. https://doi.org/10.1108/EJIM-07-2016-0066

Vogus, T. J., & Sutcliffe, K. M. (2007). Organizational resilience: Towards a theory and research agenda in 2007 IEEE International Conference on Systems. *Man and Cybernetics* (pp. 3418–3422).

Von Hippel, E. (1994). *The sources of innovation*. Oxford University Press.

Xiao, L., & Cao, H. (2017). *Organizational resilience: The theoretical Model and research implication* [Paper presentation]. ITM Web of Conferences, 12. https://doi.org/10.1051/itmconf/2017/204021

Yin, R. (2018). *Case study research and applications: Design and methods*. (6th ed). SAGE Publications Ltd.

Zook, C., & Allen, J. (2016). *The founder's mentality*. Harvard Business Review Press.